Fiscal Year:	FY 2015	**Task Last Updated:**	FY 05/27/2016
PI Name:	Hirsa, Amir Ph.D.		
Project Title:	Amyloid Fibril Formation in Microgravity: Distinguishing Interfacial and Flow Effects		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	BIOPHYSICS--Biophysics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	12180-3522	**Congressional District:**	20
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2013 Complex Fluids & Macromolecular Biophysics NNH13ZTT001N
Start Date:	09/01/2013	**End Date:**	08/31/2018
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	3	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Gorti, Sridhar	**Contact Phone:**	256 544-0158
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Lopez, Juan (Arizona State University)		
Grant/Contract No.:	NNX13AQ22G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Formation of amyloid fibrils is widely studied because they appear to play a role in numerous neurodegenerative diseases. The fibrillization process is characterized by a nucleation step followed by fibril growth, much like the steps that occur during protein crystallization. Aside from the general interest in macromolecular biophysics, recent evidence shows the relevance of amyloid fibril formation in animal models exposed to cosmic radiation, with obvious implications for interplanetary travel. A preponderance of evidence shows that agitating protein solutions can significantly accelerate amyloid fibril formation; however, the mechanisms by which the presence of bubbles and flow independently affect fibrillization are not known. The situation is further complicated because we do not yet have mechanistic models that can quantify the influence of natural convection and other effects of gravity on the fibrillization process. We know that fluid interfaces induce amyloid fibrils when they are sheared, but the influence of container walls is not known. Here, we aim to distinguish the influence of flow and fluid interfaces on amyloid formation. Formation of		

Task Description:	protein amyloid fibrils provides a model system for studying biomolecular transport phenomena, where the influence of flow can be altogether negated by the elimination of gravity. Furthermore, the experiments can be performed aboard the International Space Station (ISS) with essentially no solid container (surface tension bounded). Thus, the free surface can be readily observed (e.g., with high magnification non-oil objectives), manipulated (e.g., can be sheared or left otherwise undisturbed), and modified (e.g., by spreading a lipid bilayer). First, we will perform experiments at 1g on a small droplet to ascertain the effect of forced convection on a liquid surface. Next, we will generalize our very successful computational model in annular systems to the spherical geometry of liquids at microgravity. We will design the space-based experiments based on the results from the ground experiments and numerical predictions. Finally, we will perform experiments aboard ISS on large-scale drops using the Light Microscopy Module (LMM) facility.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	When proteins are perturbed from their native state, they become unfolded and then can bond with other proteins, forming an aggregate. This is the reason why egg becomes white and solid when cooked. Amyloid fibrils are a unique type of protein aggregate that occur naturally in the body. We study them because amyloid plaques in the brain are associated with many neurodegenerative disorders, including Alzheimer's. Fibrils are very interesting because they are highly ordered and self-assemble, so when you heat insulin, for example, it denatures and bonds with other denatured monomers (or assists in denaturing the other monomers). These form a nucleation site, and from there the fibril elongates as more monomers bond in a beta sheet structure. The primary mechanism for activation of protein in the body is not yet understood, but it is likely to be a combination of flow and interfacial effects that native proteins are exposed to. We have developed a new laser probe for in-situ turbidity measurements within the flow apparatus to get highly sensitive data, as solutions begin clear and then become turbid as amyloid aggregates form. To our knowledge this provides the first time-resolved, non-invasive measurement of turbidity in a flowing system, providing unique insight into the dynamics of flow-induced amyloid formation.
Task Progress:	Below is a list of accomplishments made during Year-2 of this NASA project. A) The concept of a knife edge setting fluid in the bulk into motion by the action of surface shear viscosity predicted computationally last year was demonstrated through experiments. The results of the computational study were published recently in an archival journal (Lopez & Hirsa 2015; see Bibliography). The experimental results along with corresponding numerical predictions were submitted recently for publication in an archival journal, Journal of Fluid Mechanics (Raghunandan, Lopez & Hirsa). B) Uniform shear generated in a rotating Couette flow device was found to significantly accelerate amyloid fibril formation primarily through increasing the nucleation rate. A maximum in nucleation rate was revealed by a two-parameter model of the kinetics for both varieties of insulin studied. The experimental results along with various kinetic models were published recently in an archival journal, Journal of Physical Chemistry B (McBride et al.; see Bibliography). C) Preliminary studies of insulin in a deep-channel surface viscometer performed at room-temperature reveal significant degree of fibril formation at time scale of hours (cf without shear, fibrillization would take a year or more). Progress is made possible by a novel combination of instruments (including Brewster angle microscope, real-time turbidity probe) to augment conventional measurements, including microplate readout of native protein concentration, ThT fluorescent measurement of amyloid formation, and AFM images of amyloid fibrils. D) Technological progress has been made on the preliminary design of the RSD. A CAD design was made by engineers at NASA Marshall for a test cell placed under the LMM. The protein solution is made into a 2.5 cm scale drop constrained between a stationary ring and a rotating ring with sharp edges. E) Preliminary modeling using a commercial code (COMSOL) has been initiated to simulate the formation of the drop and its pinning by the two constraining drops.
Bibliography Type:	Description: (Last Updated: 04/15/2026)
Articles in Peer-reviewed Journals	Young JE, Posada D, Lopez JM, Hirsa AH. "Flow-induced 2D protein crystallization: characterization of the coupled interfacial and bulk flows." Soft Matter. 2015 May 14;11(18):3618-28. http://dx.doi.org/10.1039/c5sm00429b ; PubMed PMID: 25805062 , May-2015
Articles in Peer-reviewed Journals	Lopez JM, Hirsa AH. "Coupling of the interfacial and bulk flow in a knife-edge viscometer." Physics of Fluids. 2015 Apr;27(4):042102. http://dx.doi.org/10.1063/1.4916619 , Apr-2015
Articles in Peer-reviewed Journals	McBride SA, Tilger CF, Sanford SP, Tessier PT, Hirsa AH. "Comparison of human and bovine insulin amyloidogenesis under uniform shear." Journal of Physical Chemistry B. 2015 Aug;119(33):10426–33. http://dx.doi.org/10.1021/acs.jpcb.5b04488 , Aug-2015
Dissertations and Theses	McBride SA. "Effect of Hydrophobic Residues on Amyloid Formation under Steady Shear and at Interfaces." MS Thesis, Rensselaer Polytechnic Institute, May 2015. , May-2015